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Engineering Atomically Co–N_3B/C Configuration for Rigid–Soft Coupling SEI in High-Performance Sodium-Ion Batteries

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The solid electrolyte interface (SEI) on electrode materials remains a critical challenge in the development of alkali metal-ion batteries, with its complex formation mechanism and diverse composition significantly impacting practical performance. Herein a theory-guided prediction approach is presented to construct a rigid–soft coupling SEI, engineered through the strategic regulation of both interfacial and bulk electrochemistry for the design of nitrogen- and boron-coordinated cobalt single atoms within nitrogen-doped carbon (Co–N_3B/C) anodes. The experimental and theoretical findings reveal that the distinctive Co–N_3B configuration tends to generate a local electric field that benefits for fast Na^+ transfer kinetics and the decomposition of fluorinated electrolyte, resulting in the formation of a robust inorganic-rich SEI. Consequently, the dual-coordinated Co single-atom electrode design, coupled with the hybrid organic–inorganic SEI, endows the sodium-ion batteries anode with a high capacity of 230.5 mAh g^{-1} at 2 A g^{-1} and exceptional long-term cycling stability, as evidenced by 97.0% capacity retention over 5000 cycles at 2 A g^{-1}. This study not only demonstrates the significant advantages of single-atom catalysis in controlling SEI formation and electrolyte degradation but also paves the way for using this technology to enhance the performance of carbon-based anodes in a wide range of metal-ion batteries.

1. Introduction

Sodium-ion batteries (SIBs) have garnered considerable attention for grid-scale energy storage systems, primarily because of the abundance of sodium resources. However, their commercial viability is impeded by several issues, including low initial Coulombic efficiency, inadequate rate capability, and poor cycling stability mainly due to the Na dendrite formation.[1] A major strategy to mitigate Na dendrite formation involves the modification or reconstruction of the solid-electrolyte interphase (SEI).[2] The SEI is known to have a mosaic structure composed of various organic and inorganic components, e.g., $ROCO_2Na$, Na_2CO_3, Na_2O, and NaF.[3] Notably, NaF, which has a low Na^+ diffusion barrier of 0.12 eV and satisfactory electronic insulation properties, accelerates the homogenization of Na^+ flux and ensure uniform Na deposition while preventing electrons passage through

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the SEI film, respectively.[4] However, an entirely inorganic SEI is relatively brittle, which makes it prone to stress-induced cracking that can negatively affect the long-term cycling performance of the battery.[5] Alternatively, an ether-derived SEI may consist of dense inorganic species in the thick inner layer with the thin dense organic components distributed in the outer layer. Nevertheless, ether-based electrolytes are susceptible to chemical decomposition at high applied potentials and are expensive, particularly for the Na^+ battery systems.[6] In contrast, the in situ constructed ester-SEI rarely exhibits the hybrid features of ether-SEI, typically presenting an uneven inorganic–organic hybrid structure derived from the mainstream carbonate ester-based electrolytes. This limitation further hinders interfacial Na^+ transfer kinetics, resulting in poor rate capability.[7] Consequently, there is a demand, although challenging, to construct a hybridized SEI with a NaF-rich inner layer and an organic outer layer by precisely controlling the ester-based electrolyte decomposition of ester-based electrolytes, particularly focusing on the C–F dissociation chemistry.

Researchers have developed a variety of efficient SEI layers incorporating both organic and inorganic components mainly through three main pathways.[8] First, enhancing both the homogeneity and mechanical stability of the SEI is essential to reduce ongoing reactions between metal anodes and electrolytes.[5b] Recent advancements employed polymer-based molecular protection and LiF to construct organic–inorganic composite layers.[9] However, challenges remain, including poor interface compatibility, internal agglomeration between polymer matrices and inorganic materials, energy density loss, and elevated costs due to the inclusion of an inert modified layer. Second, extensive efforts have been dedicated to electrolyte engineering to create a mosaic structure composed of diverse organic and inorganic components by incorporating fluorine-containing additives or solvents and enhancing the coordination of fluorine-containing anions.[2a,10] Unfortunately, the high cost, viscosity, and poor wettability of these methods restrict their potential for industrial applications. Consequently, both strategies are frequently implemented through complex and time-consuming post-synthetic processes. Noteworthy, the components of electrode materials play a crucial role in directing SEI formation, as they help to establish a stable electrode/electrolyte interface by controlling electrolyte decomposition kinetics, which recognized as an effective strategy.[11] Although several studies have introduced the concept of interfacial catalytic using nanostructured electrode materials to facilitate SEI formation via electrolyte reduction,[6,12] establishing a universal structure-activity relationship between the intrinsic electronic structure of doped-nanostructured catalysts and their catalytic activity remains a challenge. Thus, it is crucial to strategically modulate the electronic configuration of well-defined active sites in anode materials to precisely and uniformly catalyze the decomposition of ester-based electrolytes.

Single-atom materials have attracted much interest in heterogeneous catalysis owing to their well-defined individual metal atoms, optimized atom efficiency, and remarkable catalytic properties.[13] The carbon-based metal (M)-N_x moieties have been demonstrated to have the catalytic activity in the conversion of alkali-metal compounds or organic electrolytes.[14] This finding signifies that single-atom active sites can be employed to modulate the SEI layer of carbon-based anodes, thereby simultaneously improving the sodium storage kinetics and cycling stability of these anodes. Additionally, unlike the common M-N_4-C moiety, the local symmetry-broken metal single-atom site can be achieved by introducing heteroatoms such as S, B, and P, which possess varied atomic radii and electronegativities.[15,16] This modification can alter electronic (anti-)bonding states on specific d/s/p orbitals and the planar symmetry of the active sites, further enhancing the hybridization interactions in a rationally symmetry-reduced structure. Boron (B), an element with lower electronegativity atom and an atomic radius similar to N, has proven to be an effective electrode dopant for sodium-ion batteries and other advanced battery systems.[17] To elucidate and control the electrolyte degradation mechanism, it is highly desirable to fabricate a well-defined dual-coordinated single atom with optimized electronic properties, especially focusing on regulating the redox state of the electrolyte related to the electron transfer.

Herein, we focus on the development of electrode engineering, using density functional theory (DFT) calculations and electrochemical investigations to enables fast and ultra-stable sodium storage. We then design and fabricate the boron and nitrogen co-coordinated cobalt single atoms within the nitrogen-doped carbon (Co SAs/NBC) anodes, which can exhibit a high capacity (230.5 mAh g^{-1} at 2A g^{-1}), excellent rate capability, and cycling durability (97.0% capacity retention over 5000 cycles at 2 A g^{-1}). The Advanced characterizations and mechanism studies further confirm that Co–N_3B site of Co SAs/NC act as the active centers and catalyze the directed decomposing of $NaPF_6$ and fluoroethylene carbonate (FEC) to achieve the formation of a thin and robust rigid–soft coupling SEI with an inorganic-rich inner and fast Na^+ transfer kinetics.

2. Results and Discussion

The well-defined atomic structures of transition metal single-atom materials render them highly effective at enhancing the adsorption energies and facilitating the uniform deposition of alkali metal ions.[18] Recent studies have demonstrated that single-atom catalysts composed of transition-metal cobalt (Co) centers, coordinated by various chemical bonds, exhibit exceptional catalytic activity in several key electrochemical reactions.[19] Inspired by these findings, we first conducted in-depth DFT calculations to investigate the interfacial diffusion behaviors of Na^+ on various dual-coordinated single-atom configurations, such as Co-N_3S/C, Co-N_3P/C, Co-N_3F/C, and Co-N_3B/C. The current analysis reveals that the Na^+ migration energy barrier from the carbon lattice to the Co-N_3B center is the smallest among all four configurations, indicating that the Na^+ tends to migrate more easily toward the nearby region rather than locally accumulate (**Figure 1a**). Notably, the Co-N_3B/C site maintains constant diffusion energy as the coordination number (≥5) increases. This finding suggests that Co-N_3B/C has a much lower diffusion barrier for Na^+ compared to Co-N_3F/C, Co-N_3S/C, and Co-N_3P/C materials, highlighting its potential as a highly active site for enhancing Na^+ transfer kinetics. Furthermore, the adsorption energies (E_a) between the atomically dispersed metal site or carbon site and Na in various moieties are calculated to evaluate the anchoring ability toward Na^+. As shown in Figure 1b, Co-N_3B/C displays the highest adsorption energies for Na^+ at -1.65 eV, surpassing that of Co-N_3F/C (-0.60 eV), Co-N_3S/C (-1.27 eV), Co-N_3P/C (-1.34 eV), Co-N_4/C (-1.49

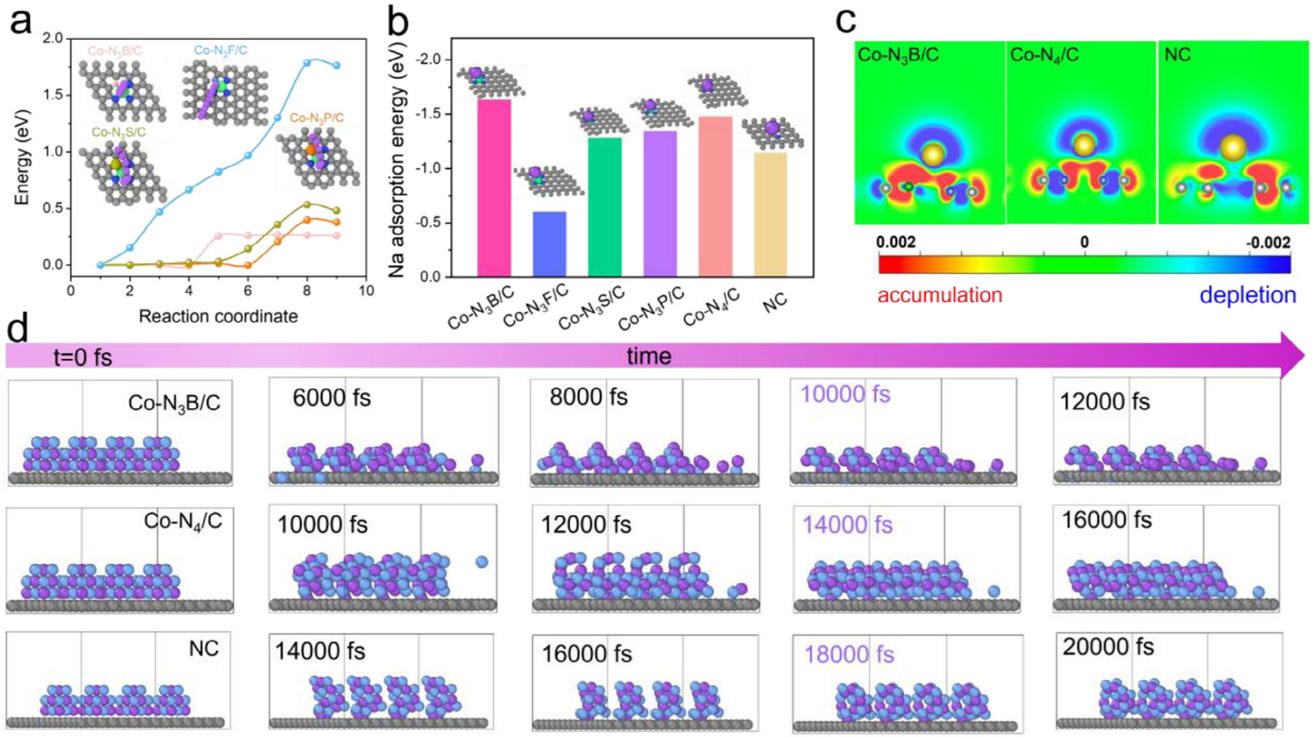

Figure 1. a) The diffusion energy barriers of Na^+ from the carbon atomic ring to the various dual-coordinated Co sites including Co-N_3S/C, Co-N_3P/C, Co-N_3F/C, and Co-N_3B/C. b) The adsorption energies of Na^+ on the various models. c) Electron localization function maps of Co-N_3B/C, Co-N_4/C, and NC. d) Featured snapshots of NaF clusters evolution and corresponding stable structures on various anode surfaces, including Co-N_3B/C, Co-N_4/C, and NC, respectively.

eV), and NC (-1.14 eV). This finding suggests that incorporating the Co-N_3B moiety into the carbon matrix significantly enhances the capacity of the material to absorb Na^+, thereby enabling it to preferentially absorb Na atoms at anode/electrolyte interface. To further explore the bonding properties of adsorbed Na^+ on NC, CoN_4, and Co-N_3B configurations, a charge density difference analysis has been conducted. This analysis visualizes the electron density depletion from Na atoms and the corresponding accumulation around different coordinated moieties, as depicted in Figures S1 and S2 (Supporting Information). Notably, the Co-N_3B/C showed a stronger affinity between Co-N_3B and Na^+ compared to NC and CoN_4 sites, indicating that the Na^+ can be effectively anchored to Co-N_3B sites. Furthermore, the Co-N_3B/C component exhibits significantly greater adsorption activity for Na^+ than NC and CoN_4, attributed to the electron-deficient effect of B.[20] This is corroborated by the calculated electron localization function maps (Figure 1c; Figure S3, Supporting Information), which align with the enhanced Na^+ storage performance revealed by the electronic density (Figure S1 and S2, Supporting Information). The results confirm that Co-N_3B/C possesses an outstanding adsorption activity and a low diffusion barrier for Na^+, which synergistically enhancing the Na^+ storage performance through more favorable interactions and charge redistribution processes. On the other hand, to better understand the growth behavior of NaF spherical clusters on three anode-based surfaces, simulated annealing-molecular dynamics (MD) simulations from machine learning potentials (MLPs) are conducted.[21] The local NaF deposition morphology at various evolution times, recorded every 2000 fs, is presented in Figure 1d and Figures S4–S6 (Supporting Information). As observed, the NaF cluster morphology on the Co-N_3B/C model evolve into a spherical nanosized particle and stabilizes at only 10 000 fs, achieving stability more quickly than Co-N_4/C (14 000 fs) and NC (18 000 fs) models. As displayed in Figure 1d, the number of interacting NaF clusters in the first solvation shell is displayed. The cluster formation energy of NaF on Co-N_3B/C is the lowest among the three anodes, indicating that NaF can be stably clustered more easily on Co-N_3B/C compared to Co-N_4/C or NC.

To confirm the theoretical predictions outlined above and further fabricate the high-performance SIBs, we proceeded with the synthesis and subsequent characterization of Co-N_3B/C configuration. The carbon-based Co-N_3B moiety (Co SAs/NBC) is successfully obtained via a typical two-step process including the precursor synthesis and subsequent carbonization process, as illustrated in **Figure 2**a, in which dimethyl imidazole served as C and N precursors, boric acid as B source and cobalt nitrate as Co precursor, respectively. Typically, Co and B sources are added with a certain stoichiometric ratio to synthesize the templates (denoted as Co,B-ZIF-8). The Co SAs/NBC was obtained by pyrolyzing the Co, B-ZIF-8 precursor under N_2 atmosphere. This pyrolysis process of Co, B-ZIF-8 is also clearly tracked by thermogravimetric analysis (TGA) (Figure S7, Supporting Information). For a thoroughly comparative study, the reference materials are further prepared with a similar method to that of Co SAs/NBC, in which the material is derived from the precursor without the B source (Co-ZIF-8) and was identified as Co SAs/NC, and the material

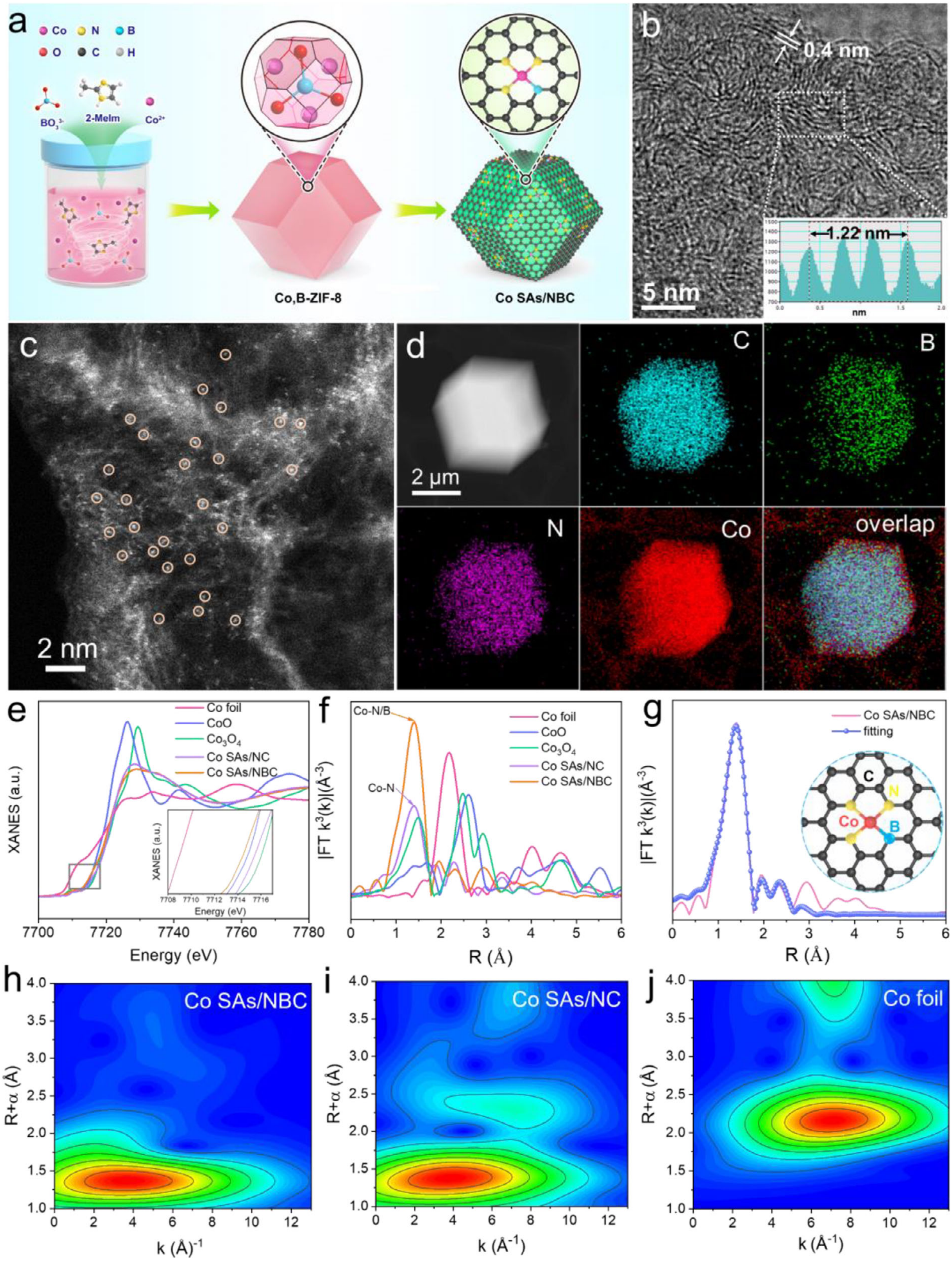

Figure 2. a) Schematic illustration for the synthetic process of Co SAs/NBC. b) High-resolution TEM and c) AC-HAADF-STEM images of Co SAs/NBC. d) EDS elemental mapping of Co SAs/NBC. e) The Co K-edge XANES and f) FT k^3-weighted EXAFS of the various samples. g) EXAFS fitting curve of Co SAs/NBC. The inset displays the sketched structure model of Co SAs/NBC. The blue, green, purple, and red balls refer to C, B, N, and Co elements, respectively. h–j) WT-EXAFS plots of Co SAs/NBC, Co SAs/NC, and Co foil.

derived from the precursor (ZIF-8) without the Co and B sources was synthesized and named as nitrogen-doped carbon (NC).

Phase characterization and composition analysis of the samples are carried out by powder X-ray diffraction (XRD) (Figure S8, Supporting Information). The XRD pattern revealed that the pristine Co,B-ZIF-8 exhibited no significant difference to ZIF-8 in the crystal pattern, which indicated the formation of Co,B-ZIF-8 instead of ZIF-67. After pyrolyzing, it presented two broad diffraction peaks at 28° and 43.8° (Figure S8b, Supporting Information), corresponding to the (002) and (101) lattice planes of amorphous carbon, respectively. It is noted that there are featureless peaks of Co-related crystal phases observed in the XRD pattern of Co SAs/NBC, which can be ascribed to the low loading amount of Co species. Significantly, compared with the Co SAs/NC, the (001) diffraction peak of Co SAs/NBC shifts to a lower angle due to the presence of C-B coordination. Additionally, the morphology and microstructure of the as-prepared various samples are characterized by scanning electron microscope (SEM) and transmission electron microscopy (TEM) measurements. The Co,B-ZIF-8 and the reference samples both possess a similar octahedral morphology with smooth surfaces (Figures S9 and S10, Supporting Information). After the pyrolysis treatment at a high temperature (950 °C), the Co,B-ZIF-8, Co-ZIF-8, and ZIF-8 are converted into Co SAs/NBC, Co SAs/NC, and NC, respectively (Figures S11–S13, Supporting Information). As depicted in Figure 2b, Co SAs/NBC displays an expanded interlayer spacing (d_{002}) of 0.4 nm, which is beneficial to the insertion/extraction of Na^+.[22]

The aberration-corrected high-angle annular dark-field scanning (AC-HAADF-STEM) images (Figure 2c; Figure S14, Supporting Information) confirms that the Co single atoms (as labeled in orange circles) are evenly distributed across/within the carbon substrate in a high loading, which was further demonstrated by the ICP-MS result. Energy-dispersive X-ray spectroscopy (EDS) mapping revealed that C, B, N, and Co elements distribute uniformly over the whole dodecahedra (Figure 2d). As shown in Figure S16 (Supporting Information) of the Raman spectrum, Co SAs/NBC shows two representative peaks at 1594 and 1349 cm^{-1}, attributed to G band and D band, respectively.[23] For the Co SAs/NBC, the intensity ratio of I_D/I_G band is 0.976, which is higher than that of Co SAs/NC (0.956) and NC (0.940), demonstrating that the structural defects of the Co SAs/NBC is induced by the Co,B co-doping. As reflected by the Brunauer–Emmett–Teller (BET; Figure S17a, Supporting Information) and pore volume (Figure S17b, Supporting Information), the Co SAs/NBC exhibits a large specific surface area (507.2 $m^2\ g^{-1}$) and porosity-rich structure, which is beneficial to improve the exposure of active sites and enhance the Na^+ transport. The X-ray photoelectron spectroscopy (XPS) was further conducted to characterize the electronic structure of the Co SAs/NBC (Figure S18, Supporting Information). As illustrated in the high-resolution Co 2p spectrum (Figure S18a, Supporting Information), the deconvolution peaks of Co SAs/NBC positioned at 795.8 and 780.5 eV are characteristic of Co $2p^{3/2}$ and Co $2p^{1/2}$, with the satellite peaks observed at 785.3 and 802.9 eV. The N 1s XPS spectrum of Co SAs/NBC (Figure S18b, Supporting Information) could be deconvoluted into four peaks positioned at 402.1, 400.5, 399.4, and 398.1 eV, corresponding to the oxidized N, graphitic N, Co–N, and pyridinic N chemical environments, respectively. The B 1s XPS spectrum (Figure S18c, Supporting Information) indicated that the Co SAs/NBC exhibits two dominant peaks at 190.7 and 188.1 eV, which are attributed to B-C and Co-B bond, respectively. Moreover, we further investigate the X-ray near-edge structure (XANES) measurements to identify the electronic structure of Co species in Co-based substrates, including Co SAs/NC and Co SAs/NBC (Figure 2e). The threshold energy of Co SAs/NBC in Co K-edge XANES spectrum is located between those of Co foil and CoO, suggesting the average oxidation state of Co ($Co^{\delta+}$) species in Co SAs/NBC,[24] which is consistent with results of XPS analysis. The Fourier-transformed (FT)-extended X-ray absorption fine structure (EXAFS) analysis of Co SAs/NBC showed a predominant peak at around 1.36 Å, attributed to the scattering of Co-N and Co-B paths in the first coordination shell, without the Co-Co bond peak at around 2.18 Å, which is consistent with the N K-edge XANES results (Figure 2f; Figure S20, Supporting Information). The coordination configuration of Co SAs/NBC is quantitatively characterized to be a pre-designed planar $Co-N_3B$ local coordination by least-squares EXAFS fitting, which is shown in Figure 2g and Table S2 (Supporting Information). The high-resolution wavelet transform-EXAFS (WT-EXAFS) in both k and R spaces is further conducted to identify the atomically dispersed state of Co species in Co SAs/NBC (Figure S21, Supporting Information). The WT counter curve of Co SAs/NBC presents a maximum signal at around 3.8 $Å^{-1}$ (Figure 2h), ascribed to the different light atom coordinator, which is different from Co SAs/NC (4.0 $Å^{-1}$) (Figure 2i) and Co foil (5.1 $Å^{-1}$) (Figure 2j).

To assess the Na^+ battery performances, cyclic voltammogram (CV) profile of a half-cell fabrication with Co SAs/NBC-incorporated electrode is performed in the potential range of 0~3 V with a scan rate of 0.1 mV s^{-1}. As shown in Figure S23 (Supporting Information), two broad irreversible peaks are observed at approximately 1.2 and 0.4 V in the 1^{st} cycle, ascribed to the decomposition of electrolytes and the subsequent generation of SEI layer, respectively. In the following cycles, the peak intensity of the sodiation curves suddenly decreases in the 2^{nd} cycle and gradually stabilizes in the 3^{rd} cycle, due to the catalysis effect of $Co-N_3B$ site to facilitate the stable and uniform SEI film on the electrode surface.[25] The initial coulombic efficiency (ICE) of 58% is obtained from the initial discharge capacity of 364.7 mAh g^{-1} and charge capacity of 640.0 mAh g^{-1} at 0.2 Ag^{-1} based on the weight of whole Co SAs/NBC (**Figure 3a**; Figures S24, and S25, Supporting Information), which is higher than that of Co SAs/NC (48.1%) and NC (36.8%). This initial capacity loss can be attributed to the irreversible formation of the SEI layer as well as decomposition of the electrolyte. And, average CE of Co SAs/NBC is closely kept around 100%, compared to that of both Co SAs/NC and NC with obvious fluctuations (Figure S25, Supporting Information). Moreover, the Co SAs/NBC demonstrates a high reversible capacity of 340.7 mAh g^{-1} after 300 cycles at 0.2 A g^{-1} with a high retention of 93.4% (Figure 3a). In contrast, the Co SAs/NC shows only 70.7% capacity retention, while the NC exhibits a retention of 58.9% after the same number of cycles. Rate performance of Co SAs/NBC in SIBs was further evaluated systematically. The Co SAs/NBC-based electrode exhibits the highest rate capabilities of 352.9, 307.3, 274.1, 251.4, 230.5, 203.4, and 169.6 mAh g^{-1} at 0.1, 0.2, 0.5, 1, 2, 5, and 10 A g^{-1} (Figure 3b), respectively, which also outperformed those of the state-of-the-art carbon-based electrodes (Figure 3c).[26] Even by increasing the high current rate to 15 A g^{-1}, Co SAs/NBC still

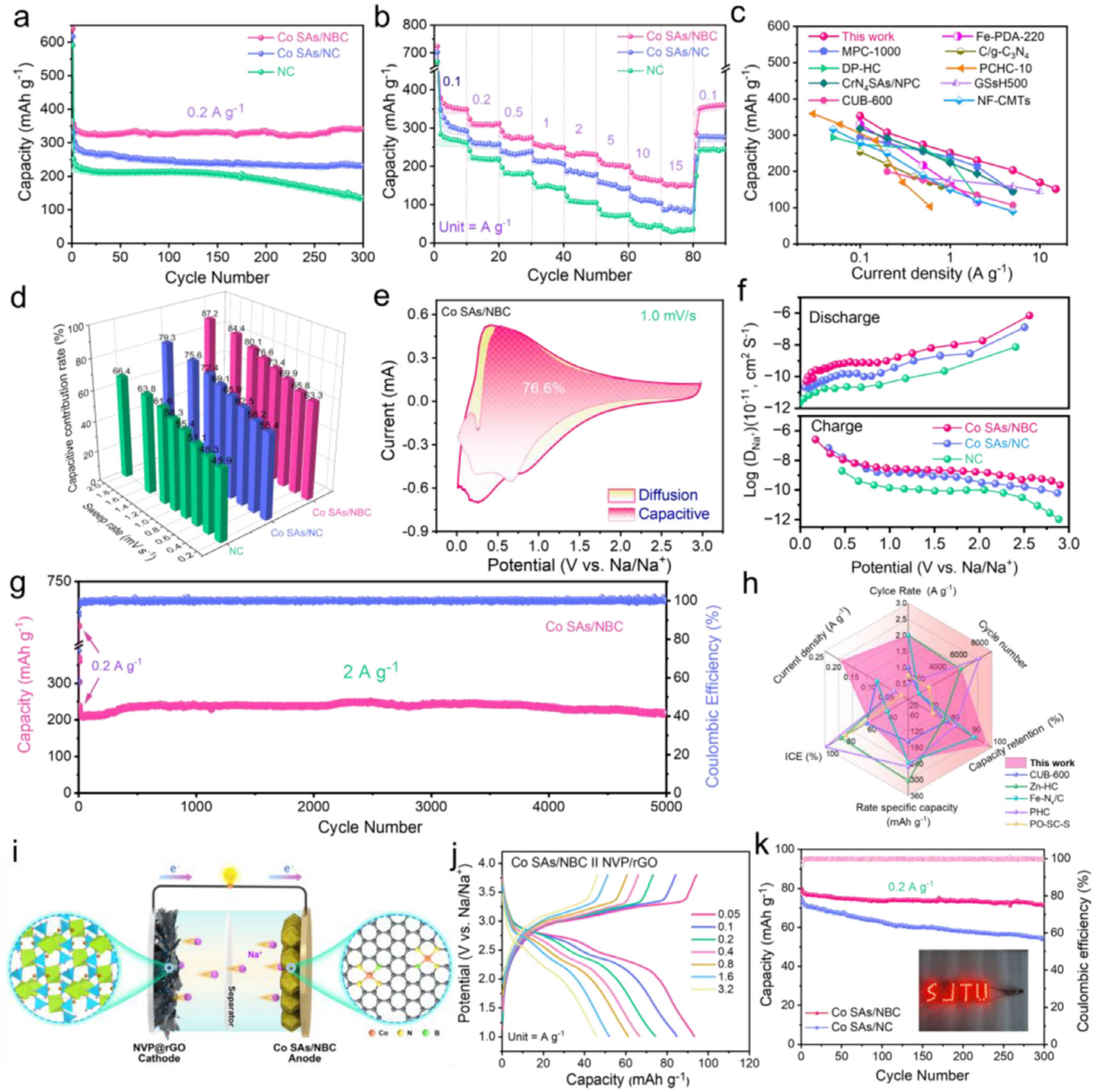

Figure 3. a) The capacity comparison of Co SAs/NBC, Co SAs/NC, and NC at 0.2 A g^{-1}. b) Rate capabilities of various anodes from 0.1 to 15 A g^{-1}. c) Comparison of rate capability with the typical anodes reported previously. d) Normalized contribution ratio of capacitive capacities at various scan rates and e) capacitive contribution of Co SAs/NBC at 1.0 mV s^{-1}. f) Na$^+$ diffusion coefficients of Co SAs/NBC, Co SAs/NC, and NC. g) Long-term cycling stability of Co SAs/NBC electrode at 2 A g^{-1} for 5000 cycles. h) Comparison of the overall performance of Co SAs/NBC with those reported and advanced electrodes in Na-ion battery. i) Schematic illustration of the Na-ion full cell with Co SAs/NBC||NVP/rGO couple and j) corresponding charge/discharge curves at 0.05 to 3.2 A g^{-1}. k) Cycling performance of Co SAs/NBC and Co SAs/NC anodes-based full battery at 0.2 A g^{-1}. Inset of photograph shows that an Co SAs/NBC||NVP/rGO full cell can light up LED lights.

delivered a remarkable reversible capacity of 151.4 mAh g^{-1}, which was much higher than the value of Co SAs/NC (84.5 mAh g^{-1}) and NC (36.4 mAh g^{-1}). When the current density is witched back to 0.1 A g^{-1}, it can still maintain a high reversible capacity of 351.7 mAh g^{-1} (Figure 3b). Since the rate capability is simultaneously determined by the electronic and ionic conductivity of the electrode, the outstanding rate property of the Co SAs/NBC anode reflects the boosted Na$^+$ transfer kinetics by the SEI layers.

To scrutinize the sodium ion storage mechanism and investigate the intrinsic reaction kinetics of Co SAs/NBC, the CV measurement is conducted at various scan rates (Figure 3d;

Figure S26, Supporting Information). As the scan rate step-wise increases to 1.0 mV s^{-1}, the Co SAs/NBC can reach 76.6% of the total capacity contribution during the capacitance-controlled process (Figure 3e), which is significantly larger than that of Co SAs/NC (69.1%) and NC anodes (58.3%) (Figure S26d,e, Supporting Information), manifesting that this electrochemical process is determined by the pseudocapacitive behaviors. With increasing the scan rate in, the electrochemical reactions more favorably occur on the surface of electrode materials in a capacitive-controlled Na-storage manner due to the requirements of fast mass transport and electron transfer, which induces an increase of capacitive contribution.[22] This result can be ascribed to promote the electric conductivity, enrich the structural defects and electrochemically active sites caused by the Co SAs/NBC, demonstrating the "adsorption-insertion" sodium storage mechanism and fast Na^+ transfer kinetics.[22] The CV curves recorded at higher scan rates display good shape consistency, expect for that the intensities of redox peaks and the surrounding area gradually enhance with an increment of scan rate, which further signifies a high-rate capability of Co SAs/NBC electrode.[27] In addition, the galvanostatic intermittent titration technique (GITT) (Figure S27, Supporting Information) and electrochemical impedance spectra (EIS) (Figure S28, Supporting Information) also confirm that the Co SAs/NBC possesses more favorable Na^+ diffusion and faster charge transfer than that of Co SAs/NC and NC (Figure 3f). In consideration of the effect of the optimized/modified anode/electrolyte interface on cycling performance, the cycling stability was systematically studied under different current densities. Even as cycled at a high current density of 2 A g^{-1}, a capacity retention of 97.0% (a capacity decay of 0.001%) with an average CE (≈100%) is demonstrated upon 5000 cycles (Figure 3g). The capacity of the Co SAs/NBC-based battery increases gradually up to approximately 400 cycles at a high current density of 2 A g^{-1}, which can be attributed to the increased interlayer distance of carbon support and enhanced surface wettability between electrode materials and liquid electrolyte, which can provide more active sites for Na^+ storage.[28] Overall, at a lower or higher current density, Co SAs/NBC still delivers an excellent cycling stability. The greatly enhanced cycling performance results from the advanced interfacial components and chemical stability, owing to suppressed irreversible side reactions at the anode/electrolyte interface after interfacial modification. The spider chart showed the superior performances consisting of Co SAs/NBC anode in comparison with some state-of-the-art SIBs systems (Figure 3h).[12a,25,26h,29] To assess the practical merit of Co SAs/NBC anode in Na^+ battery, a full cell was fabricated by integrating this composite anode with $Na_3V_2(PO_4)_3$/rGO (NVP/rGO) cathode (Figure 3I; Figure S29, Supporting Information). Moreover, the full cell also exhibits a superb rate performance with the specific capacities of 91, 85, 75, 65, 61, 52, and 45 mAh g^{-1} at 0.05, 0.1, 0.2, 0.4, 0.8, 1.6, and 3.2 A g^{-1}, respectively (Figure 3j; Figure S30, Supporting Information). As seen in Figure 3k, the prototype battery could deliver a higher discharge capacity of ≈80 mAh g^{-1} with an excellent capacity retention of 93.2% after 300 cycles at 0.2 A g^{-1}, superior to those of Co SAs/NC (71 mAh g^{-1}; 77.0%). The ICE in the full-cell system of Co SAs/NBC||NVP/rGO is 96.5% at 0.2 A g^{-1}. Totally, the achieved remarkable overall energy performance demonstrated that the as-fabricated Co SAs/NBC||NVP/rGO system possesses excellent kinetics and superior stability (Figures S31 and S32, Supporting Information).

To disclose the influence of the SEI on Na^+ storage behavior, we systematically investigate the chemical structure and composition of the SEI on various anodes after the first fully discharge applying time of flight secondary ion mass spectrometry (TOF-SIMS), XPS, and cryogenic–transmission electron microscopy (cryo-TEM) characterizations. TOF-SIMS analysis is first conducted to investigate composition distribution of SEI. **Figure 4a,b** show the depth profiles of the typical negative species in the components of SEI. As shown in Figure 4a, the NaF_2^- fragment (inorganic species), from the PF_6^- and FEC, is ascribed to NaF; The organic fragments including CH_3O^-, $C_3H_3O^-$, and C_2HO^- are also detected.[7b,29a] Significantly, no discernible signals are detected in the NaF_2^- spectra at the surface of SEI formed in Co SAs/NBC-systems, and obvious signal began to emerge after 3 s etching. In contrast, the intensity of NaF_2^- for the SEI in Co SAs/NBC during the etching time of 3s-~400s is much stronger than that in Co SAs/NC (Figure 4b), demonstrating the inorganic-rich SEI structure of Co SAs/NBC anode. Moreover, the intensity of NaF_2^- of Co SAs/NBC exhibited a higher value than that of Co SAs/NC during the etching time of ≈400 s, which indicate that this etching depth nearly cover the overall SEI, consistent with the representative depth of B signal from TOF-SIMS result (Figure S33, Supporting Information). The depth values of the NaF_2^- and C_2HO^- for Co SAs/NBC are less than those of the Co SAs/NC, which implies that the Co SAs/NBC-induced SEI is thinner than the Co SAs/NC-induced SEI. Combining the Figure 4c,d, it indicates that the inorganic compound is abundant throughout the SEI in the Co SAs/NBC anode, while the organic compounds are dominated in the outer surface of SEI.

The XPS depth profiling coupled with Ar-ion sputtering is employed on the various anodes to ascertain the compositional content of the SEI. The O1s peaks of the SEI on the Co SAs/NBC anode, appearing at 536.3, 533.4, 531.4, and 529.9 eV, are linked to the inorganic compounds containing F–O, O–C, O = C, and O–Na, respectively, primarily resulting from the decomposition of the $NaPF_6$-solvent complex (Figure 4e).[30] The C1s peaks of Co SAs/NBC anode-based SEI at 284.8, 286.2, 286.9, 288.4, and 289.3 eV are attributed to C-C, C-O, C≐O, CO_3^{2-}, and C-F, respectively; These organic components arise from solvent reduction (Figure 4f). The SEI surface in the Co SAs/NBC system exhibits a pronounced peak at 687.5 eV (P-F) and a smaller peak at 684.5 eV (Na-F) in the F 1s spectrum (Figure 4g), which originate from the reduction of PF_6^- and FEC.[29a] The intensity of the P-F peak in the SEI consistently decrease with the increased etching depth, whereas the intensity of F-Na peak at 684.5 eV significantly increased after 5 nm of etching and remained elevated. This observation indicates that inorganic components dominate the inner SEI layer, whereas the organic components reside on the outer layer. Furthermore, with prolonged sputtering, the NaF signal remain robust, suggesting that NaF-rich SEI is uniform, dense, and resilient. Nevertheless, the ratios of inorganic species in the SEI on the Co SAs/NC anode are significantly lower than those on the SEI of Co SAs/NC and NC anodes, especially for NaF (Figures S34 and S35, Supporting Information). This finding suggests that $NaPF_6$ or FEC is prone to be selectively catalyzed into NaF rather than $Na_xPO_yF_z$ or Na_xPF_y by the Co SAs/NBC anode. Additionally, these inorganic-rich components help to

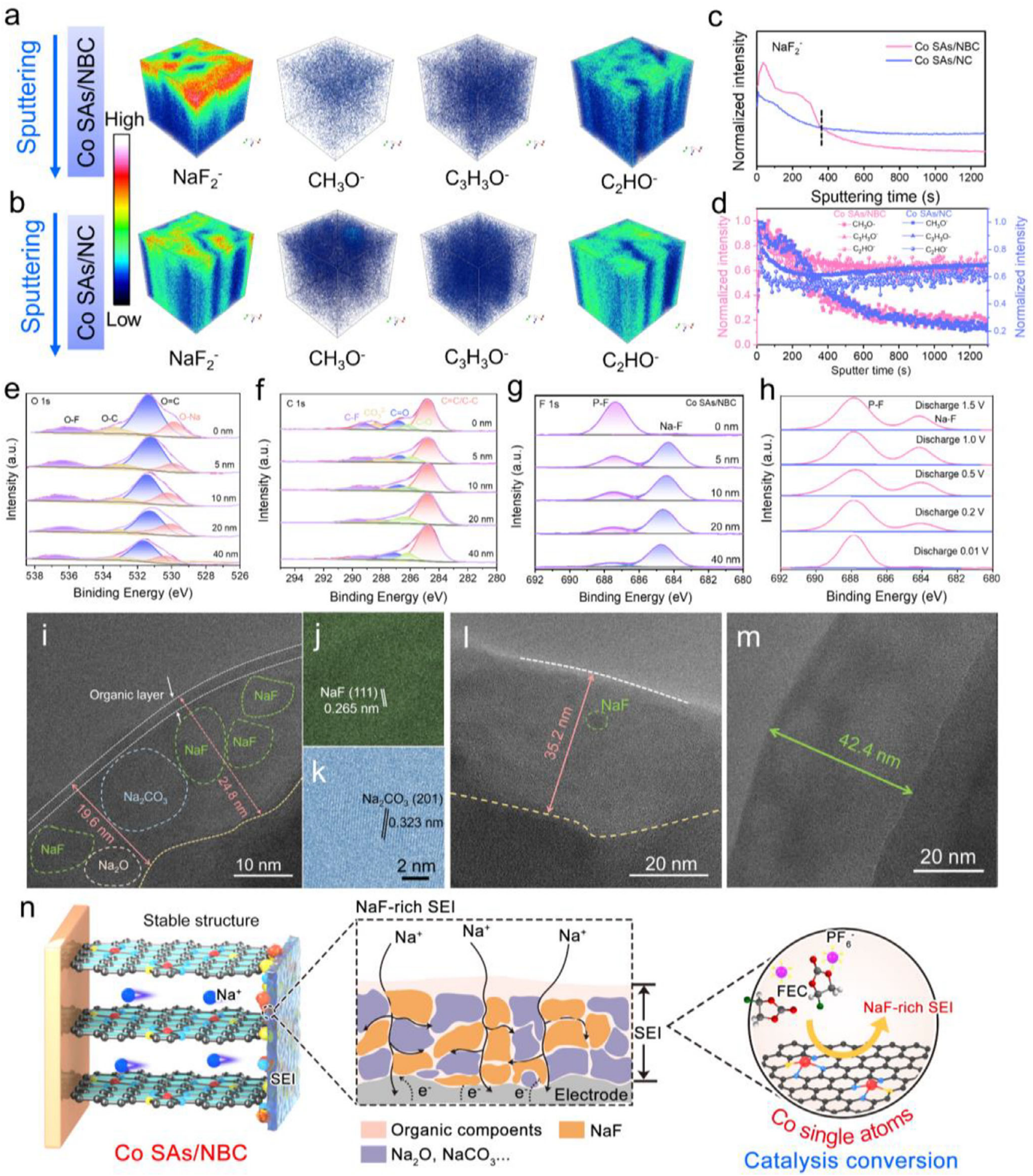

Figure 4. Overlaid 3D renderings for the secondary ion fragments NaF_2^-, CH_3O^-, $C_3H_3O^-$, and C_2HO^- on Co SAs/NBC a) and Co SAs/NC b), respectively. c) Secondary-ion yields versus sputtering time (s) for the inorganic secondary ion fragments (NaF_2^-). d) Secondary-ion yields versus sputtering time (s) for the organic secondary ion fragments: CH_3O^-, $C_3H_3O^-$, and C_2HO^-. High-resolution O 1s e), C 1s f), and F 1s g) XPS spectra of Co SAs/NBC anode with increasing the sputtering thickness. h) Ex situ F 1s XPS spectra of Co SAs/NBC anode at different discharged potentials of the initial cycle. Cryogenic TEM images of the SEI components on Co SAs/NBC i–k), Co SAs/NC l), and NC m) anodes. n) Mechanism schematic of the SEI formation via the single-atom catalysis of electrolyte reduction by the Co SAs/NBC.

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construct a mechanically stable SEI, exhibiting better uniformity and higher Na^+-conductivity, thereby enhancing to stabilization of the Co SAs/NBC-based anode and significantly improving its cycling and rate performances.

The cryo-TEM is further employed to directly observe the microstructure of the SEI. As depicted in Figure 4i, the SEI on the Co SAs/NBC anode displays smoother surfaces with dense structure, indicating a consistent development of the SEI during its formation. The SEI layers are characterized by large crystalline regions contain significantly large-areal crystal regions in the inner layer and very thin amorphous regions in the outer layer, suggesting an inorganic-dominated and layered structure, which is similar to that SEI derived from ether electrolyte.[31] The lattice spacings correspond to Na_2CO_3, NaF, and Na_2O (Figure 4i–k). In Figure 4l,m, a few distinct crystalline regions (inorganic components) with a limited thickness and loose distribution are observed, suggesting that the SEI on the Co SAs/NC or NC anodes is organic-dominated due to significant solvent decomposition, which well matches with the XPS results. The thin and inorganic-rich SEI layer on the Co SAs/NBC electrode aligns with the low and stable R_{SEI} value (Figure S28, Supporting Information) and the stable local graphite structure, which facilitates rapid Na^+ ion storage kinetics and improve rate performance. Compared to the SEI thickness on Co SAs/NC (35.2 nm), NC anodes (42.4 nm), and NBC (36.9 nm) (Figure S36e, Supporting Information), the controlled catalysis by Co SAs/NBC results in the formation of an efficient and ultrathin SEI (aroud 20 nm), significantly reducing the Na^+ transfer distance and enhancing sodium storage performance. The findings implies that the Co-N_3B site acts as an active center for modulating the kinetics of inorganic species formation within the SEI. Additionally, the introduction of rapid SEI formation via a NaF-rich layer can effectively curb the excessive decomposition of electrolytes.[32] In contrast, the Co SAs/NC, NC, and NBC anodes feature a higher concentration of organic species in the outer and thicker SEI layer, which is well consistent with the XPS depth profiling results. Such ultrathin and inorganic-rich SEI could inhibit the continuous electrolyte decomposition and minimize the undesirable side reactions, thereby promoting rapid charge transfer and superior mechanical properties. This leads to enhanced cyclic stability and improved rate performance. Overall, the precisely defined single-atom catalysis of preferential salt reduction, combined with the guiding effect of the dual-coordinated site, which collectively facilitates the development of a multifunctional organic/inorganic-rich composite solid-electrolyte interphase in ester electrolytes (Figure 4n).

To gain systematic understanding of the sodiation/desodiation reaction mechanism and structural changes of Co SAs/NBC electrode during the charge/discharge process, in situ Raman and ex situ XAFS analyses have been investigated. The intensity of both the D and G bands are gradually weakened during the sodiation process (**Figure** 5a,b), which can be attributed to the introduction of Na^+ occupying the defective sites and nanopores as well as the formation of the Na─C bond. Upon the sodiation/desodiation process, the slowly increasing I_D/I_G ratio at low potentials during the discharge and the slowly decreasing I_D/I_G ratio at high potentials during the charge combinedly suggest that the surface defect adsorption does not induce significant structural change (Figure 5c,d). With the continuing discharge to the plateau voltage of 2.88–0.03 V, the G-band shift from the frequency of 1604 cm^{-1} to 1595 cm^{-1}, while the G-band shifts to a higher frequency of around1604 cm^{-1} in the subsequent desodiation process, confirming the robust structure and highly reversible processes during the charging–discharging process of the SIBs (Figure 5c; Figure S37, Supporting Information).[33] In the XANES spectrum of the Co K-edge (Figure 5e), the pre-edge peak intensity with the sodiation state (marked with the gray dashed arrow) is higher than that of the pristine Co SAs/NBC due to the insertion of sodium ions, manifesting a decreasing number of unoccupied d-states for Co species upon reduction. During the subsequent charging, the absorption intensity of pre-edge peak edge gradually increases until it almost coincides with the absorption edge of the pristine Co SAs/NBC in the charging state, indicating that the number of unoccupied d-states for Co species increase. A small peak of Co-Na (located at 2.24 Å) signal appears during the discharge state compared to the pristine state of the pristine Co SAs/NBC due to the insertion of sodium ions,[34] and the absorption edge gradually disappear in the subsequent charging process (Figure 5f; Figure S38, Supporting Information), indicating that the local Co sites show a good reversibility during the sodiation/desodiation process, which agrees well the excellent stability during the discharge and charge process (Figure S39, Supporting Information).

To elucidate the catalytic mechanism of B,N dual-coordinated Co single atoms in promoting the formation of NaF, the DFT calculations are employed. The decomposition evolution of the $NaPF_6$ molecule in a mixture of ethylene carbonate, propylene carbonate, and FEC solutions, as obtained from the ab initio molecular dynamics (AIMD) simulations. The computations reveal that FEC and $NaPF_6$ decomposed into NaF in the Co SAs/NBC system more rapidly than when catalyzed by Co SAs/NC or NC (Figure 5g). Figure 5h illustrates a multiple electrocatalytic mechanism at various electrode-electrolyte interfaces, selectively catalyzing the breaking of C–F bonds and driving the rapid formation of the NaF.[6] Based on the results, we can conclude that Co SAs/NBC effectively reduces the energy required for C─F bond disintegration and promotes the decomposition of fluorinated compounds, while facilitating the linkage of Na^+ with electron-rich F to form NaF (Figure 5i). To provide a clearer understanding of binding energy between NaF and Co-N_3B/C, a detailed calculation of the integrated crystal orbital Hamilton population (ICOHP) values for these three cases revealed that the bonding strength follows the order: Co-N_3B/C > Co-N_4/C > NC (Figure S40, Supporting Information). This suggests that during the nucleation process, Co initially plays a critical role in interacting with fluorinated species, while the presence of B further strengthens the anchoring effect for F, thereby facilitating NaF nucleation. These findings demonstrate that Co SAs/NBC effectively catalyzes the directed decomposition of FEC, resulting in the formation of a NaF-rich SEI. The NaF inorganic species have an electronically insulating and low diffusion barrier for Na^+, which can effectively prevent electrons from entering the SEI and boost the Na^+ diffusion kinetics, further enabling the fast and efficient Na^+ storage.[35]

3. Conclusion

Using a catalytic Co-N_3B configuration-based anode, we have developed a strategy to regulate and construct a multifunctional

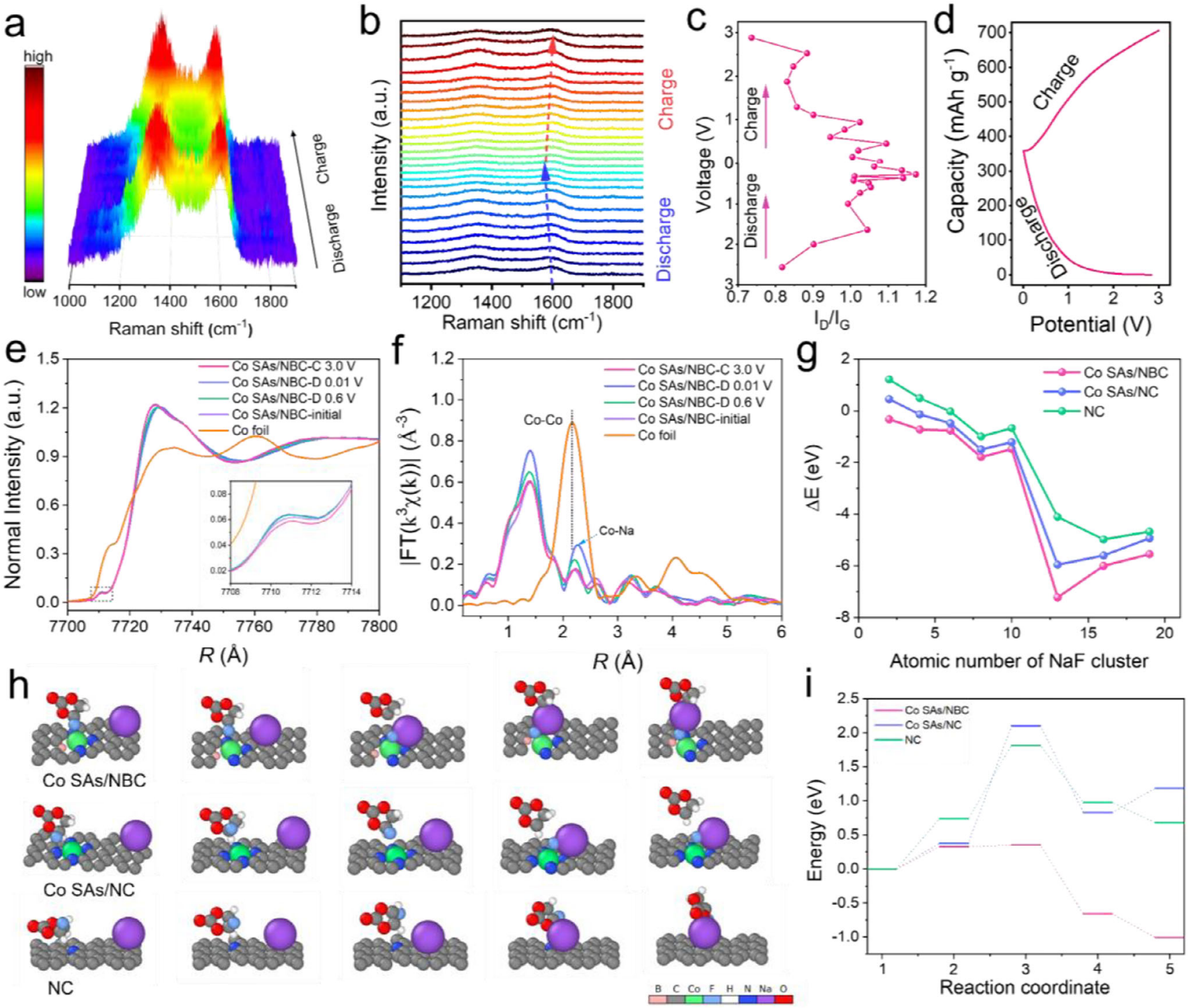

Figure 5. a) 3D projection In situ Raman spectra of Co SAs/NBC and b) corresponding In situ Raman spectra of Co SAs/NBC during the initial discharging and charging processes. c) The corresponding potential vs. I_D/I_G and d) the discharge-charge curve at the current density of 0.1 A g^{-1} during the in situ Raman measurement. e) Ex situ Co K-edge XANES spectra and f) Ex situ Co K-edge EXAFS spectra of Co SAs/NBC during the first discharged and charged states. g) The NaF formation energy of the various anodes along the number of NaF cluster. h) Proposed multiple electro-catalytic mechanisms on the various electrode-electrolyte interfaces. i) The reaction activation energies of the NaF formation at different anodes.

NaF-rich SEI. The theoretical predictions reveal that the N,B-dual coordinated Co single site features an optimized charge structure and tunable electron localization. This configuration enhances Na^+ absorption, accelerates the decomposition of the C–F bond, and reduces the energy barrier for NaF formation. Based on these predictions, we synthesized Co single atoms on NBC with a unique Co-N_3B unit using a host–guest strategy, confirmed through a series of characterizations including AC-HAADF-STEM, XAFS, and XPS. When utilized as an anode for sodium-ion battery, it exhibits competitive performance, with a specific capacity of 230.5 mAh g^{-1} at 2 A g^{-1} and a cycle stability of 97.0% capacity retention over 5000 cycles at 2 A g^{-1}. This performance is further validated in a full cell, demonstrating high capacity at 0.2 A g^{-1}, which indicates a significant potential for Na-ion battery device applications. Additionally, the formation of NaF is attributed to the synergistic electrocatalytic effect of Co SAs/NBC, as demonstrated by systematic experimental results (TOF-SIMS, depth-dependent XPS, and Cryo-TEM) and theoretical calculations. This study highlights the interplay between single-atom site catalysis and Na^+ migration kinetics that will aid in guiding the controlled design of efficient artificial SEI layers.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

anodes, rigid-soft coupling strategy, single-atom catalysis, sodium-ion batteries, solid electrolyte interphase

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